Effect of Hole Geometry on the Thermal Performance of Fan-Shaped Film Cooling Holes

Michael Gritsch
Will Colban[1]
Heinz Schär
Klaus Döbbeling

ALSTOM (Switzerland) Ltd.,
Brown Boveri Str. 7,
Baden 5401, Switzerland

This study evaluates the impact of typical cooling hole shape variations on the thermal performance of fan-shaped film holes. A comprehensive set of experimental test cases featuring 16 different film-cooling configurations with different hole shapes have been investigated. The shape variations investigated include hole inlet-to-outlet area ratio, hole coverage ratio, hole pitch ratio, hole length, and hole orientation (compound) angle. Flow conditions applied cover a wide range of film blowing ratios $M=0.5$ to 2.5 at an engine-representative density ratio $DR=1.7$. An infrared thermography data acquisition system is used for highly accurate and spatially resolved surface temperature mappings. Accurate local temperature data are achieved by an in situ calibration procedure with the help of thermocouples embedded in the test plate. Detailed film-cooling effectiveness distributions and discharge coefficients are used for evaluating the thermal performance of a row of fan-shaped film holes. An extensive variation of the main geometrical parameters describing a fan-shaped film-cooling hole is done to cover a wide range of typical film-cooling applications in current gas turbine engines. Within the range investigated, laterally averaged film-cooling effectiveness was found to show only limited sensitivity from variations of the hole geometry parameters. This offers the potential to tailor the hole geometry according to needs beyond pure cooling performance, e.g., manufacturing facilitations. [DOI: 10.1115/1.2019315]

Introduction

In state-of-the-art gas turbine cooling schemes film cooling is a well-accepted and widely used means to keep the surface temperatures of components with high thermal loads like first stage vanes and blades at acceptable limits. Coolant air is released through so-called film holes on the surface and is supposed to form a protective layer on the surface that protects the surface from hot gas exposure. Recent improvements in the manufacturing process have led to a greater flexibility in choosing a proper shape for these holes. Standard film cooling schemes have used cylindrical holes but increasing attention is given to holes with a diffuser shaped exit (so-called fan-shaped holes) due to superior thermal performance compared to cylindrical holes. The flow deceleration in the diffuser section of the hole reduces the momentum of the coolant jet and thus the penetration of the coolant into the hot gas flow. The coolant stays attached to the surface and forms an effective film. Additionally, the diffuser provides an improved lateral spreading of the coolant and thus a more uniform film on the surface since neighboring cooling jets start merging immediately downstream of the injection location

Although film holes with noncylindrical exit shape have received increasing attention during the last years, there are only very few studies that try to find out which are the governing geometrical parameters that really affect the thermal performance of the hole. Most of the studies focused on a comparison of fan-shaped hole performance versus a standard cylindrical cooling hole [1–3] or on a comparison of global film cooling schemes employing cylindrical and/or fan-shaped holes, e.g., comparing a double row of cylindrical holes vs. a single row of fan-shaped holes [4,5] or combining rows of cylindrical and fan-shaped holes [6].

Gritsch et al. [7], Thole et al. [8], Gritsch et al. [9–11] compared the performance of two different fan-shaped holes in terms of film-cooling effectiveness and heat transfer coefficients. They found that the thermal performance of a hole that diffuses in both lateral and streamwise direction (laidback fan-shaped hole) is superior to a hole that only diffuses in lateral direction, in particular at high blowing ratios. The area ratio of both holes were almost identical ($AR=3.0$ for the lateral diffusing hole and $AR=3.1$ laidback fan-shaped hole). Saumweber et al. [12] investigated a row of holes that had identical shapes as those of Gritsch et al. at elevated turbulence levels. They found that the fan-shaped and the laidback fan-shaped hole performances were very close. The same type of hole shapes have also been investigated for leading edge cooling applications [13]. The performance of the laidback fan-shaped hole was found to be superior to the fan-shaped hole for this case.

Bell et al. [14] provided film effectiveness data for a row of fan-shaped holes diffusing only in lateral direction and a row of holes diffusing only in forward direction. Since the area ratios of their fan-shaped holes are only in the order of 1.65 the thermal performance of both shaped hole geometries was found to be very close to standard cylindrical holes.

Kohli and Bogard [15] compared two fan-shaped holes featuring the same lateral diffusion angle but different forward diffusion angles of 15 and 25 deg. The hole with the larger forward diffusion angle was expected to keep the coolant jet closer to the surface. However thermal performance of this hole in terms of film cooling effectiveness was found to be lower than for the hole with the smaller forward diffusion angle due to flow separation in the diffuser part of the hole.

Lutum et al. [16] studied film-cooling effectiveness for a row of holes diffusing in both lateral and forward directions as compared to holes diffusing in lateral direction only. Hole shapes were very close to those from Gritsch et al. They report that the performance of the hole with an additional forward diffusion performed slightly

[1]Currently at the Mechanical Engineering Department, Virginia Polytechnic Institute and State University, Blacksburg, VA 24061.

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Table 1 Test rig operation conditions

Main flow	Temperature	T_t	340 K
	Pressure	P_t	1.0 bars
	Mach number	Ma	0.3
	Reynolds number	Re_D	22,000
	Turbulence level	Tu	4%
	BL thickness at film injection	δ/D	0.5
Coolant (carbondioxide)	Pressure	P_t	<1.4 bars
	Temperature	T_t	295 K
	Density ratio	DR	1.7
	Blowing ratio	M	0.5,…,2.5

worse and attribute this behavior to local flow separations within or just downstream of these laid back shaped holes.

Numerical studies [17–19] carried out on fan-shaped film hole geometries suggest that the flow field structure inside the film hole is highly complex. The hole geometrical parameters have an major impact on the flow field inside and at the exit of the hole and, consequently, on the thermal performance of the film hole.

The findings from previous studies underline the importance of in-depth knowledge of the impact of individual hole geometrical parameters. It is obvious that compared to a standard cylindrical film hole, the number of geometrical parameters that define the hole shape is profoundly increased. Beside the parameters that define a cylindrical hole [hole diameter, hole length, and the orientation of the hole axis on the surface—typically referred to as inclination and orientation (or compound) angle] there are further parameters that basically describe the shape of the diffuser-shaped exit portion of the hole. These include the starting point of the diffuser as well as the diffuser angles in streamwise and lateral direction. With those parameters fixed, the area ratio of the hole, and—with a given hole-to-hole pitch—the coverage ratio are set. These two typically serve as the standard parameters defining a fan-shaped film hole. When a film cooling design is done these parameters can be used to tune the thermal performance. However, it is crucial to know the impact of each of the dominating hole geometry parameters on the thermal performance.

Furthermore, it needs to be considered that the manufacturing process of these fan-shaped holes has a profound impact on the hole shape. For laser-drilled holes, it is cost efficient to use the same drilling process parameters for a complete row of film holes. This can cause a change in hole shape since the wall thickness of a blade typically varies in radial direction. Also blade-to-blade casting tolerances lead to different hole shapes on different blades. Moreover, tolerances in the manufacturing process can cause major hole-to-hole shape variations. When defining a manufacturing process, it is therefore extremely helpful to know which hole geometry parameters have a big impact on the thermal performance and which do not. Ultimately, those having a big effect need to be tightly controlled.

Since there are obviously no studies reported in the open literature that can provide detailed insight this study aims to assess the effect of a number of geometrical parameters of typical fan-shaped holes on thermal performance. Detailed film-cooling effectiveness distributions and discharge coefficients are used for evaluating the thermal performance of a row of fan-shaped film holes. An extensive variation of the main geometrical parameters describing a fan-shaped film-cooling hole is done to cover a wide range of typical film-cooling applications in current gas turbine engines.

Due to the large body of data, this study concentrates on the film-cooling effectiveness data. For an overall assessment of the film-cooling performance, also heat transfer coefficients need to be considered [20–22]. Those have been measured in parallel and will be presented in a follow-up paper.

Experimental Apparatus and Procedure

Wind Tunnel and Film-Cooling Injection System. The present investigation is carried out in a continuous flow wind tunnel delivering main flow air—up to 1 kg/s—into a test section of 50 mm × 130 mm cross section. An electrical 34 kW heater increases the main flow temperature to about 340 K. After passing through a flow settling device and some flow straightening grids and honeycombs the flow is accelerated in a bell-mouth shaped nozzle to Mach numbers in the order of 0.3. A turbulence grid located about 45 D upstream of the film-cooling ejection location created a turbulence intensity level of 4% at the ejection location. The length of the test plated mounted to the test section is 800 mm which allows for measuring the film-cooling effectiveness data up to 110 D downstream of the ejection location. The test plates consist of a polyurethane foam featuring a low thermal conductivity of about 0.1 W/mK providing nearly adiabatic conditions for the experiments. Each of the test plates features a row of fan-shaped film-cooling holes consisting of five holes. The metering diameter of these holes is 4 mm. The holes are fed from a 130 × 70 mm plenum underneath the test plate. The coolant flow rate provided to the coolant plenum is directly measured by means of a gas meter.

Despite that the present study focuses on the effect of hole geometry parameters, it is made sure that the main flow parameters, e.g., the blowing ratio and the density ratio are kept in engine-representative ranges. Carbondioxide is used as coolant for the experiments in order to achieve enginelike coolant-to-main flow density ratios of about 1.7. Table 1 summarizes the operating conditions.

Measurements of Film Effectiveness and Discharge Coefficients. Surface temperature measurements are carried out by means of an infrared thermography system (FLIR ThermaCAM 3000). The sensor of this system is detecting radiation in the 8 μm to 9 μm range. The IR camera is mounted above the test section of the wind tunnel (Fig. 1). The wall opposite to the test plate is equipped with four circular KCl windows that are

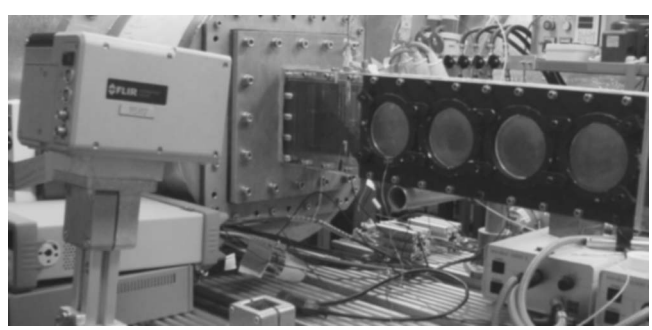

Fig. 1 Test rig setup

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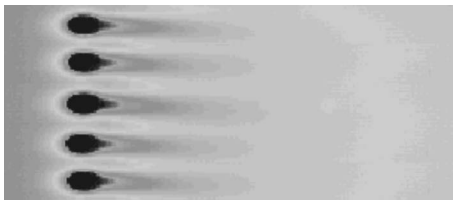

Fig. 2 Local surface temperature distribution downstream of film ejection location, Config.D

highly transparent to infrared radiation (τ=0.93) in the wavelength range of the sensor. The radiation collected from the test plate is digitized into an array of 240×320 pixels. Accounting for the optical setup used a spatial resolution of 0.3 mm×0.3 mm per pixel is achieved. The camera is mounted on a rigid stand that could slide in the streamwise direction allowing for images to be taken through each of the four windows for every test case.

The test plate surface is covered by black paint with a known emissivity of 0.95. A number of K type thermocouples are distributed on the test plate surface which are used for an *in situ* calibration of the thermography system in order to increase the accuracy of the surface temperature measurements and, thus, of the film cooling effectiveness data, following the approach described by Martiny et al. [23]. The local film-cooling effectiveness is then based on the local surface temperature, coolant supply total temperature, and the main flow recovery temperature

$$\eta = \frac{T_{aw} - T_{m,\mathrm{rec}}}{T_{c,t} - T_{m,\mathrm{rec}}} \tag{1}$$

Before applying this equation, the surface temperature readings have been corrected for back-side losses by heat conduction through the test plate material which have been estimated employing a thermocouple measuring the back-side temperature of the test plate. Due to the low thermal conductivity of the test plate material and the rather thick test plate, back-side heat losses are found to be small (less than a 2% effect on film-cooling effectiveness). Furthermore, the heat radiation from the test plate to the opposite wall and—through the windows—to the lab surroundings have been estimated. Since the experiments are carried out at a main flow temperature level only about 40 K above ambient temperature, radiation effects are found to be small (less than 1% effect on film-cooling effectiveness). A typical local surface temperature distribution is given in Fig. 2.

In order to quantify the flow losses, discharge coefficients have been measured as well. The discharge coefficient is defined as the ratio of real and ideal mass flow through the film hole.

$$C_D = \frac{m_{c,\mathrm{real}}}{m_{c,\mathrm{ideal}}} \tag{2}$$

The ideal mass flow is defined by assuming isentropic, one-dimensional expansion across the hole from coolant total pressure to main flow static pressure level.

$$m_{\mathrm{ideal}} = A_c p_{t,c} \left(p_m/p_{t,c}\right)^{(\kappa+1)/2\kappa} \sqrt{\frac{2\kappa}{(\kappa-1)RT_{t,c}}\left[\left(p_{t,c}/p_m\right)^{(\kappa-1)/\kappa} - 1\right]} \tag{3}$$

The crosssection of the initial cylindrical hole portion is used as the metering area A_c.

$$A_c = \frac{\pi}{4} D^2 \tag{4}$$

The calculation of the blowing ratio

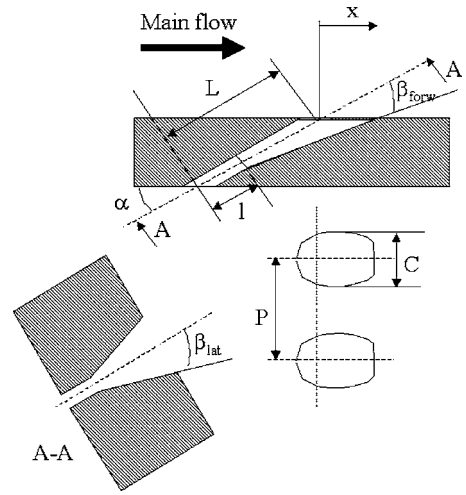

Fig. 3 Film-cooling hole geometry

$$M = \frac{m_c/A_c}{(\rho u)_m} \tag{5}$$

is based on the hole inlet cross-sectional area as well.

Experimental Uncertainties

Experimental uncertainties are estimated following the partial derivative method of Moffat [24]. Accounting for the individual uncertainties in setting the flow (main flow Mach number δMa =0.01, blowing ratio δM=0.02, and density ratio δDR=0.02) and measuring temperatures (δT=0.4 K), the relative overall uncertainty in film-cooling effectiveness is estimated to be in the range 1.5% to 9%. The lower value corresponds to locations with good film protection (η=0.7) while the higher value is valid for locations with poor cooling (η=0.05). The uncertainty in discharge coefficients ranges from 1.2% to 4.2% for the conditions applied during this study.

Film-Cooling Configurations

An initial assessment was carried out to find out which geometrical quantities describing the shape of a film-cooling hole can be easily controlled/changed and/or are prone to manufacturing uncertainties. Those of this list that are suspect to have a big impact on the film-cooling performance have been selected for this study. These include

- The length of the hole, L/D.
- The hole exit-to-inlet area ratio, AR.
- The hole pitch, P/D.
- The hole coverage (hole exit width-to-pitch ratio), C/P.
- The hole compound angle, γ.

Note that the hole inclination angle α, although having a major impact on film-cooling performance, is kept constant at 30 deg for all test cases presented in this paper. The length of the cylindrical inlet portion is ℓ=2D for all hole geometries investigated (Fig. 3).

A test matrix of 16 different hole geometries has been created. These 16 geometries can be pooled into subsets; each subset is meant for investigating a single geometrical parameter from the list above. The challenge has been to single out these effects since typically if one parameter is changed others are also affected. Therefore, the test matrix and the hole geometries associated have been carefully selected to allow for differentiating the impact of individual geometrical hole parameters. For instance, Configs A to C are supposed to investigate the effect of the length of the fan-shaped hole portion. Note that all other geometrical quantities in Table 2 below are kept constant. This implies, that Configs A to C

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Table 2 Film-cooling hole geometries investigated

	Config	L/D	AR	P/D	C/D	β_{lat} (deg)	β_{forw} (deg)	γ (deg)
	A	11.5	3.5	6	0.49	6	2	0
L/D effect	B	9.5	3.5	6	0.49	7	4	0
	C	7.5	3.5	6	0.49	7	11	0
	D	11.5	4.2	4	0.65	4	8	0
P/D effect	E	11.5	4.2	6	0.43	4	8	0
	F	11.5	4.2	8	0.32	4	8	0
	G	11.5	2.5	6	0.31	2	4	0
	H	11.5	2.5	6	0.37	4	2	0
C/P effect	I	11.5	2.5	6	0.39	4	0	0
	E	11.5	4.2	6	0.43	4	8	0
	K	11.5	4.2	6	0.57	7	2	0
	L	11.5	4.2	6	0.63	9	0	0
	M	11.5	3.5	6	0.43	4	4	0
AR effect	E	11.5	4.2	6	0.43	4	8	0
	L	11.5	4.2	6	0.63	9	0	0
	N	11.5	4.7	6	0.63	8	3	0
	E	11.5	4.2	6	0.43	4	8	0
Compound angle	J	11.5	4.2	6	0.43	4	8	45
	O	11.5	4.2	6	0.43	4	8	60
	P	11.5	4.2	6	0.43	4	8	45[a]

[a]One-side diffusion hole.

do not feature identical diffuser portions with just the length of the diffuser portion varying but are rather individual designs in order to make sure that the area ratio and the coverage ratio are kept unchanged.

The parameter ranges were chosen according to what is typically found in engines rather than covering the full feasible range. For example, the area ratio AR was varied from 3.5 to 4.7. Lower values are of course feasible, but the objective was to get a sensitivity of the film effectiveness with area ratio within the typical range applied in engines. Some of the parameters investigated, e.g., the hole length, might be more typical for land-based gas turbines than for aero engines.

Experimental Results

Discharge Coefficients. In order to quantify the flow losses associated with injection of the coolant flow into the main flow discharge coefficients have been measured for all hole configurations investigated. Measured discharge coefficients are presented in Fig. 4 for blowing ratios $M=1.0$ to 2.5. Generally, the discharge coefficients were found to be in the 0.8 to 0.9 range—decreasing with increasing blowing ratio since the diffusion and the associated pressure recovery gets less effective at higher flow rates. The pressure recovery is somewhat affected by the hole diffuser shape. However, differences from configuration to configuration are rather small—typically less than 10%. The discharge coefficients agree well with those found by Hay and Lampard [25] and Gritsch et al. [26] et al. who both reported C_D values of 0.8 for hole geometries and flow conditions that are not identical but similar to those applied in the present study.

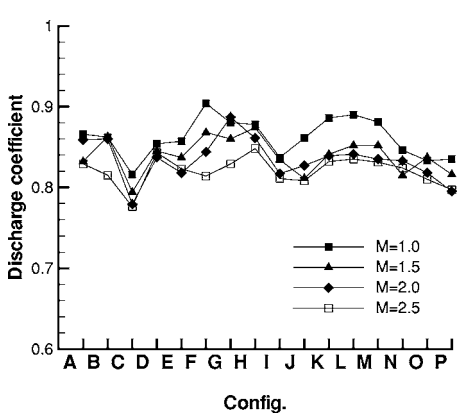

Fig. 4 Discharge coefficients for all holes investigated

Film-Cooling Effectiveness. In order to benchmark the experimental setup and procedure, experimental film-cooling effectiveness data are compared to results from previous studies (see Table 3). The comparison is made based on laterally averaged film-cooling effectiveness. Only studies employing engine-like density ratios DR are considered.

The data of Config D (P/D=4) are used in order to compare them in particular to data of Saumweber et al. [6] and Dittmar et al. [5] (both P/D=4) who present data in the far field of the injection location up to $x/D=50$. The results (Fig. 5) reveal excellent agreement with the data of Dittmar et al. [5] and—in the near field—with the data of Gritsch et al. [7]. Kohli and Bogard [15] tested hole geometries with a smaller P/D=3 and consequently found higher laterally averaged film-cooling effectiveness. Although Lutum et al. [16] employed a larger pitch of P/D=6 film effectiveness is higher than those of the present study. Their investigation was carried out on a convex surface which is known to enhance film-cooling effectiveness for low blowing rates and thus to counterbalance the effect of the larger pitch. Overall, it can be stated that the present data agree well with data from previous investigations bearing in mind that the hole shapes and operating conditions differ to some extent from study to study.

Effect of Length-to-Diameter Ratio. Lutum and Johnson [27] found that for cylindrical film holes the hole length-to-diameter ratio L/D has basically no effect on the film cooling effectiveness for L/D>5. The present study extends this finding to fan-shaped

Table 3 Literature data sets considered for comparison with current study

	M	P/D	DR	Tu (%)	α (deg)
Dittmar et al. [5]	0.5	4	1.3	9	45
Gritsch et al. [7]	0.5	4[a]	1.85	1.4	30
Saumweber et al. [6]	0.5	4	1.75	5.1	30
Lutum et al. [16]	0.5	6	1.78	5	30
Kohli and Bogard [15]	0.5	3	1.6	0.2	55
Current study, Config D	0.5	4	1.7	4	30

[a]Single-hole data are converted to a "virtual pitch" of P/D=4 as given in [6].

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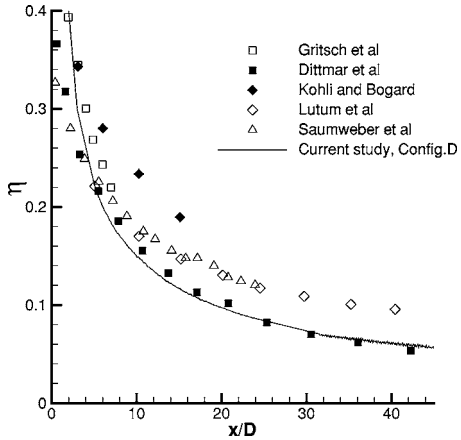

Fig. 5 Laterally averaged film-cooling effectiveness, comparison to literature data sets

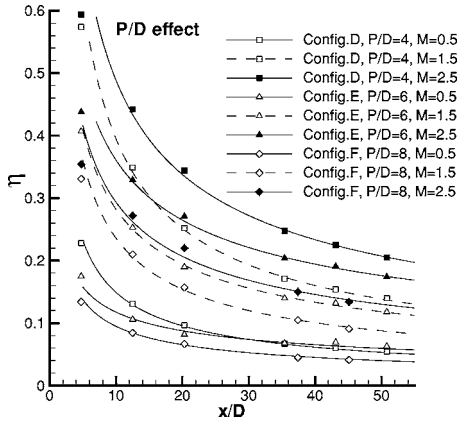

Fig. 7 Effect of hole-pitch-to-diameter ratio on laterally averaged film-cooling effectiveness

hole geometries. For the three L/D ratios investigated during this study (L/D=7.5, 9.5, and 11.5) the film-cooling effectiveness was found to be essentially the same, see Fig. 6. Since the cylindrical inlet section length was kept constant for all Configs. A, B, and C, it can be stated that the length of the hole diffuser section does not have an impact on the film effectiveness. Note that—as already mentioned in the previous chapter—Configs. A, B, and C feature the same coverage and the same area ratio. The diffuser section of each of these holes has been individually designed to meet these criteria.

Effect of Pitch-to-Diameter Ratio. The pitch-to-diameter ratio P/D is a measure for the number of holes per unit span and—for a given blowing ratio—the amount of cooling per unit span. Therefore, it is no surprise that the hole configuration Config D (P/D =4) provides a higher film-cooling performance as compared to config E (P/D=6) and F (P/D=8), Fig. 7. However from a cooling performance point of view what is the optimum P/D ratio? Since film cooling effectiveness is often assumed to scale with the reciprocal of the P/D ratio. Figure 8 presents the laterally averaged film-cooling effectiveness normalized with D/P.

If the film-cooling effectiveness would solely depend on the amount of coolant spent per unit span the curves for the different configurations should collapse. The data for P/D=6 and 8 are indeed in good agreement (except in the vicinity of the injection location) while the data for P/D=4 are somewhat lower. Apparently, for P/D=4 the holes are too close to each other to make efficient use of the coolant spent.

Effect of Hole Coverage. The hole coverage C/P is defined as the ratio of hole exit width and hole pitch which basically defines the width of the coolant film emerging from a film hole. Frequently, this parameter is assumed to describe the potential of the coolant to spread out laterally and form a self-contained film with the coolant emerging from its neighbor holes. Data have been measured for three rather low C/P ratios (0.31, 0.37, 0.39) for AR=2.5 (Fig. 9), and three rather high C/P ratios (0.43, 0.57, 0.63) for AR=4.2 (Fig. 10). Since the AR ratio is kept constant, low C/P ratios indicate that the hole is mainly forward diffusing while for high C/P ratios the hole is mainly laterally diffusing. Surprisingly, for $M=0.5$ and 1.5 all data collapse to a single curve, showing that the C/P ratio does not have an impact on the film-cooling effectiveness for these blowing ratios. For $M=2.5$, however, film-cooling effectiveness increases with C/P, indicating that the lateral diffusion is somewhat more effective than the forward diffusion.

Effect of Area Ratio. The area ratio AR is defined as the ratio of cross-sectional areas at hole exit and inlet. This parameter is assumed to describe the potential of the hole to diffuse the coolant in the fan-shaped portion of the hole and, thus, to reduce the momentum of the coolant. Lower momentum leads to less penetration into the main flow and consequently higher film cooling performance. Data have been measured for three area ratios (3.5, 4.2, and 4.7). Since for a fair comparison the coverage needs to be kept constant, two area ratios (3.5 and 4.2) were investigated for C/P=0.43 (Fig. 11), and another two area ratios (4.2 and 4.7) for C/P=0.63 (Fig. 12). Since the C/P ratio is kept constant, high AR

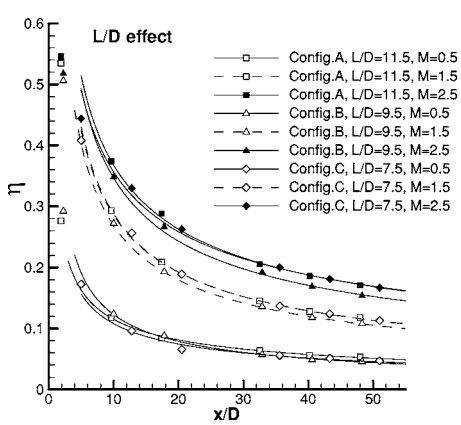

Fig. 6 Effect of hole-length-to-diameter ratio on laterally averaged film-cooling effectiveness

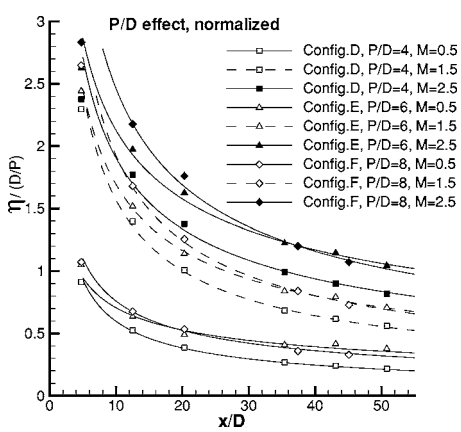

Fig. 8 Effect of hole-pitch-to-diameter ratio on laterally averaged film-cooling effectiveness, normalized

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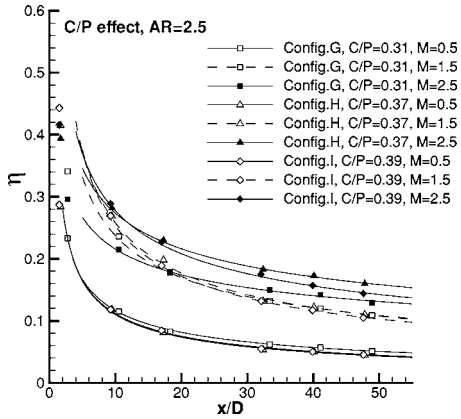

Fig. 9 Effect of coverage ratio on laterally averaged film-cooling effectiveness at AR=2.5

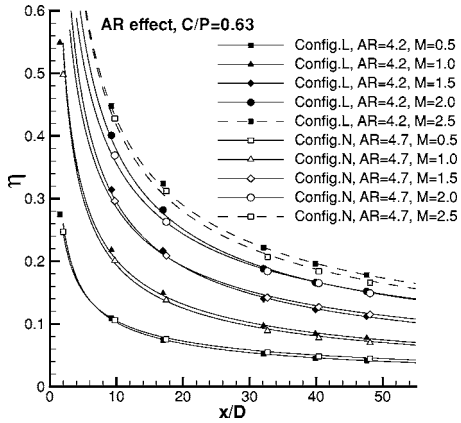

Fig. 12 Effect of area ratio on laterally averaged film-cooling effectiveness at C/P=0.63

ratios indicate that the hole is more forward diffusing than for a lower AR ratio. As for the C/P parameter, all data were found to collapse indicating that the AR ratio does not have an impact on film-cooling performance within the range investigated. The selected AR ratios cover what is typically found on engine airfoils. It should be noted though that the results may look different for lower AR ratios than investigated in this study.

Effect of Compound Angle. For cylindrical film holes, adding a compound angle is known to improve the lateral spreading of the coolant and reduce the penetration of the coolant jets into the main flow. This results in enhanced film-cooling effectiveness data, but typically at the expense of higher heat transfer coefficients due to intensified interaction of coolant and main flow.

Whereas cylindrical holes with compound angles have been investigated for years, it was just recently that some results on compound shaped holes were published.[5,14,28–30] Although poor lateral spreading or penetration into the main flow are generally not an issue for fan-shaped holes, putting a compound angle on a fan-shaped hole is beneficial under certain circumstances and has thus received increasing attention. Therefore, the current study is looking into the effect of adding a compound angle to a fan-shaped hole on film-cooling effectiveness. Three different compound angles (γ=0 deg, 45 deg, and 60 deg) have been investigated (Fig. 13). At low to medium blowing ratios (M=0.5, 1.5), adding a compound hole seems to have no major impact on film-cooling effectiveness, while at high blowing ratios (M=2.5) the performance of the holes featuring a compound angle is somewhat lower than for a straight hole. The difference between a 45 deg and 60 deg is negligible. Apparently, putting a compound angle leads to elevated interaction of the noncoflowing coolant and main flow which reduces the protective effect of the film.

One option to overcome this problem is to change the diffuser portion of the hole in a way that the diffuser is not symmetric anymore but employs a one-sided diffusion. This means that the lateral diffusion angle on the windward side of the diffuser is smaller than the one on the leeward side $\beta_{\mathrm{lat},w} < \beta_{\mathrm{lat},l}$ (Fig. 14). This would cause the flow leaving a hole to be directed more in the main flow streamwise direction. Figure 15 shows a compari-

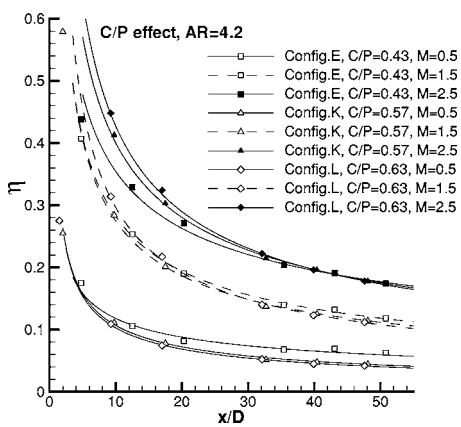

Fig. 10 Effect of coverage ratio on laterally averaged film-cooling effectiveness at AR=4.2

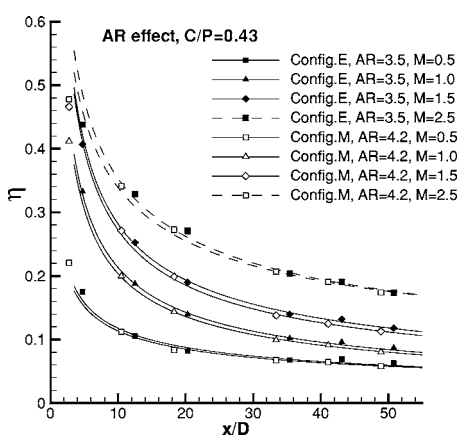

Fig. 11 Effect of area ratio on laterally averaged film-cooling effectiveness at C/P=0.43

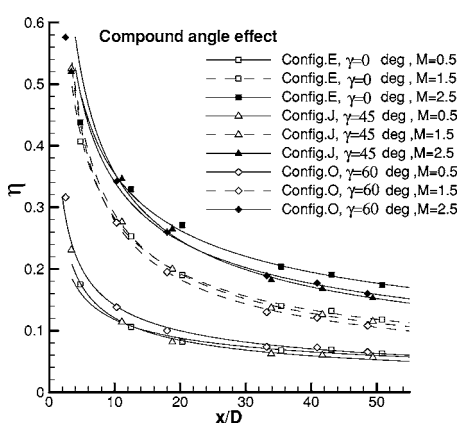

Fig. 13 Effect of compound angle on laterally averaged film-cooling effectiveness

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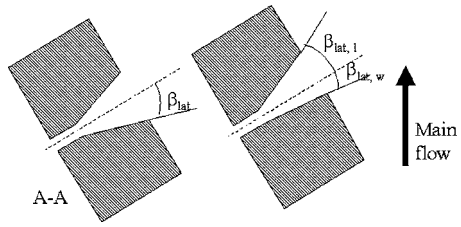

Fig. 14 Symmetric (left) and nonsymmetric diffuser portion of the film-cooling hole

son of a hole with a symmetric diffusion to a hole with a one-sided diffusion. Apparently, there is some benefit in terms of film-cooling effectiveness that can be gained by reshaping the hole at medium to high blowing ratios.

Summary

An experimental study has been carried out to investigate the impact of various hole geometry parameters on the discharge coefficient and film-cooling effectiveness of fan-shaped film-cooling holes. This study is unique since a detailed compendium of all major hole parameters is provided. The key findings may be summarized as follows:

- Pressure losses quantified by means of discharge coefficients are not profoundly affected by the geometry of the fan-shaped hole. All C_D values were found to be in a 0.8 to 0.9 range.
- Within the range investigated, most hole geometry parameters only have a weak impact on laterally averaged film-cooling effectiveness.
  (1) The effect of hole length to diameter ratio is small.
  (2) The effect of pitch-to-diameter ratio is as expected large as it determines the amount of coolant spent per unit span.
  (3) The effect of coverage ratio and area ratio is basically negligible.
  (4) Putting a compound angle has some detrimental affect at high blowing ratios which can be overcome by using non-symmetric diffusion.
- Since many of the parameters investigated do only have a small impact on filmcooling effectiveness the cooling designer may be more tolerant on the hole shape if requirements from manufacturing process necessitate to do so

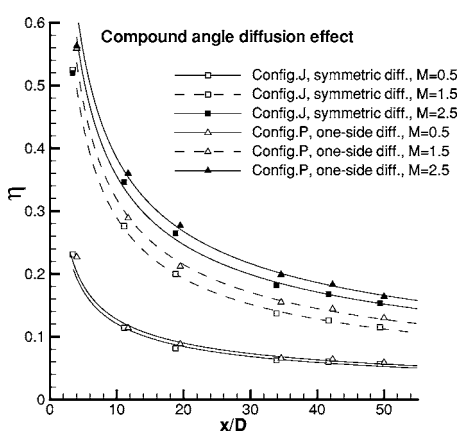

Fig. 15 Effect of compound angle diffusion on laterally averaged film-cooling effectiveness

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Nomenclature

AR = Hole inlet-to-exit area ratio
C_D = Discharge coefficient
C = hole width, exit side, mm
D = Film-cooling hole diameter, mm
DR = Density ratio
I = momentum flux ratio
L = Film-cooling hole length, mm
l = Length of hole cylindrical inlet portion, mm
Ma = Mach number
M = Blowing ratio
m = Mass flow rate through the film-cooling hole, kg/s
p = Static pressure, Pa
p_t = Total pressure, Pa
P = hole-to-hole pitch, mm
R = Gas constant
Re_D = Reynolds number based on film hole diameter
T_t = Total temperature, K
Tu = Turbulence intensity, %
α = Angle of hole inclination (deg)
β = Hole diffusion angle, deg
δ = boundary layer thickness, m
γ = Angle of hole orientation (compound angle), deg
κ = Ratio of specific heats
η = Laterally averaged film-cooling effectiveness
ρ = Flow density, kg/m3

Subscripts

c = Internal flow conditions ("coolant")
m = External flow conditions ("mainflow")
rec = Recovery conditions
AW = Adiabatic wall conditions
lat = lateral (spanwise) diffusion
forw = forward (streamwise) diffusion
l = leeward side
w = windward side

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